



Byungjun Kim

M.S. Student
Graduate School of AI
Korea Advanced Institute of Science and Technology,
Daejeon, South Korea

+82 10-5001-9430

bjkim@kaist.ac.kr

www.linkedin.com/in/byungjunk

Project Page

EDUCATION

- **Korea Advanced Institute of Science and Technology, Daejeon** 2025 ~ Present
M.S., Graduate School of AI
Bio-Imaging Signal Processing Lab (BISPL)
- **Korea Advanced Institute of Science and Technology, Daejeon** 2019 ~ 2025
B.S., School of Electrical Engineering GPA: 4.12/4.3 (Summa cum laude)

ACADEMICS

- **Vector Institute (Advisor: Rahul G. Krishnan)** Sep. 2026 ~ Dec. 2026
Internship Toronto, Canada
 - Adaptive Conditioning Agentic AI for generative models
- **NeurIPS 2026 Reviewer**

EXPERIENCE

- **Barreleye Inc.** Jan. 2024 ~ Aug. 2024
Internship Seoul, South Korea
 - **Development of BI-RADS Compliant Breast Lesion Identification Algorithms**
Engineered morphological features and optimized machine learning models, achieving high accuracy in shape classification by quantifying tumor irregularities.
 - **Lightweight Deep Learning Architecture Design & Medical Device Integration**
Designed a resource-efficient backbone network using Squeeze-and-Excitation layers and Atrous Convolution Blocks, successfully deploying the prototype onto a medical device for real-time inference.

PUBLICATIONS

- **MotionCFG: Boosting Motion Dynamics via Semantic Motion Sharpening** *NeurIPS (2026)*
Byungjun Kim, Soobin Um[†], Jong Chul Ye[†]
 - Proposed **MotionCFG**, a training-free inference-time guidance method for Text-to-Video (T2V) generation that selectively perturbs motion-related concept embeddings and steers generation away from the resulting motion-degraded condition.
 - Formulated the method as **Semantic Laplacian Sharpening**, which sharpens the score landscape along motion-relevant semantic directions, and introduced an early-step piecewise guidance schedule to improve motion dynamics while preserving spatial fidelity.
 - Validated the method on state-of-the-art T2V models including Wan2.1 and CogVideoX, showing consistent improvements in motion dynamics with negligible overhead, and further extended the same principle to fine-grained semantic control such as object numerosity.
- **Diverse Text-to-Image Generation via Contrastive Noise Optimization** *ICLR (2026)*
Byungjun Kim, Soobin Um*, Jong Chul Ye[†]*
 - Spearheaded the development of **Contrastive Noise Optimization (CNO)**, a novel inference-time framework that mitigates mode collapse in diffusion models by optimizing initial noise vectors via an objective min/max interpretation.
 - Designed a contrastive loss objective operating in the denoised space to maximize semantic diversity while preserving fidelity, validated through theoretical analysis of mutual information bounds.
 - Achieved **State-of-the-Art (SOTA)** performance on the quality-diversity Pareto frontier across various backbones (SDXL, SD3, FLUX) with minimal computational overhead.

AWARDS & SCHOLARSHIPS

- **Academic Awards**
Dean's List in Spring 2021, KAIST
Dean's List in Fall 2023, KAIST
- **Scholarships**
National Excellence Scholarship (Natural Sciences and Engineering)

FIELDS OF INTEREST

- **Vision-Language-Action Models:** Multimodal Robot Learning, Action Generation, Robot Manipulation
- **Video Generation:** Autoregressive Video Generation, Text/Image-to-Video Synthesis, Long-Horizon Temporal Modeling, Motion Dynamics
- **Inference-Time Guidance in Generative Models:** Training-Free Sampling, Classifier-Free Guidance (CFG), Noise/Embedding Optimization, Adaptive Guidance Scheduling